

Enhancing Critical Thinking: Interactive Cognitive Offload Instruction with Generative AI In English Essay Writing

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Abstract

Enhancing Critical Thinking (EFL) writers often find it difficult to find a balance between linguistic precision, rhetorical depth, and procedural requirements, which compromise critical thinking. An Interactive Cognitive Offload (ICO) framework is proposed in this paper, which uses generative AI as a tool for strategically assigning mechanical writing tasks. A mixed-method study comparing ICO students and controls showed that ICO significantly improved critical thinking skills, especially in the depth of argument and evidence synthesis. The analysis showed that ICO was effective in reducing extraneous cognitive load (32%, $p < .001$) while increasing germane cognitive load (28%, $p = .004$), and there was no significant difference between groups — allowing students to focus on higher-order thinking. In addition to the positive correlation between increased germane load and these critical thinking dimensions ($r = .59-.62$, $p < .001$) and evidence synthesis ($r = -.65$, $p < .001$), there was strong negative correlation between increased germane load and these critical thinking dimensions ($r = .59-.62$, $p < .001$), confirming a “cognitive reinvestment” mechanism. Qualitative findings identified key patterns in agency negotiation: strategic delegation, reflection triggered by AI outputs, and critical assessment of AI proposals. In order to address potential limitations, the results were improved by adding cognitive load measurements to real-time observation records, as well as multiple validation strategies for thematic analysis. These results indicate that ICO framework transforms generative AI into a metacognitive framework for writing critical thinking. Implications suggest that the ICO framework effectively reduces the risk of cognitive surrogacy through structured collaboration and provides a viable model for using artificial intelligence to augment basic intellectual skills.

CCS Concepts

• **Applied computing** → Education; Computer-assisted instruction.

Keywords

critical thinking, cognitive offloading, generative AI, English essay writing

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1 INTRODUCTION

Generative artificial intelligence (GenAI) tools, exemplified by platforms like ChatGPT, present transformative opportunities for democratizing writing proficiency, offering unprecedented support in text generation and refinement. However, their rapid integration into academic contexts simultaneously poses critical challenges to the intellectual development of learners, particularly concerning the cultivation of essential critical thinking skills. Unchecked implementation risks fostering cognitive surrogacy, a concerning trend where core thinking processes are systematically delegated to machines, potentially eroding the rigor of argumentation and the authenticity of student voice [1]. This tension is acutely manifested within the domain of English as a Foreign Language (EFL) pedagogy. Learners grapple with a profound cognitive trilemma: the concurrent and demanding burden of mastering linguistic accuracy (grammar and syntax), achieving rhetorical depth (complex argumentation and sophisticated evidence synthesis), and adhering to stringent procedural requirements (citation and formatting conventions). Empirical evidence underscores the severity of this challenge, indicating that EFL students’ drafting time is consumed by grammatical concerns alone. This disproportionate focus forces detrimental trade-offs, where higher-order critical thinking is consistently compromised to accommodate lower-order mechanical demands [2], hindering the development of the very skills central to academic success.

To address this critical challenge and harness the potential of GenAI while mitigating its risks, this research introduces the Interactive Cognitive Offload (ICO) framework. Grounded in the robust theoretical foundations of Cognitive Load Theory (CLT) [3] and Agency Theory [4], ICO proposes a pedagogical paradigm shift. It advocates for the strategic redistribution of cognitive resources during the complex task of academic writing. The core principle involves intentionally leveraging GenAI to manage extraneous-load tasks, thereby freeing up scarce working memory capacity. This deliberate offloading allows learners to reserve their cognitive resources for germane cognitive processes essential to critical thinking, including argument mapping, evidentiary evaluation, logical structuring, and nuanced synthesis. Consequently, the ICO framework repositions GenAI from a potentially substitutive tool



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that might undermine intellectual engagement to a metacognitive scaffold designed to enhance learning.

2 LITERATURE REVIEW

2.1 Cognitive Load Theory and Writing Demands

Cognitive Load Theory (CLT) provides a foundational lens for understanding the cognitive challenges inherent in academic writing, particularly for EFL learners. CLT distinguishes three types of cognitive load: intrinsic (task complexity), extraneous (non-essential processing), and germane (schema construction). Academic writing imposes a triple cognitive burden on EFL students: mastering linguistic accuracy (grammar, syntax), developing rhetorical sophistication (argumentation, evidence synthesis), and adhering to procedural conventions (citation, formatting) [5]. Empirical studies reveal that EFL learners spend 68% of drafting time addressing grammatical concerns alone, leaving scant cognitive resources for higher-order thinking, a phenomenon termed the cognitive trilemma [6]. This aligns with Skulmowski & Xu's (2022) findings that excessive extraneous load impedes expertise development, suggesting that strategic offloading of mechanical tasks could optimize cognitive resource allocation [7].

2.2 Generative AI in Writing Pedagogy

Generative AI (GenAI) tools like ChatGPT offer unprecedented capabilities for text production, yet unguided implementation risks fostering cognitive surrogacy, the substitution of human intellectual effort with machine output [8]. Early research revealed contradictory outcomes: while Lee & Low (2024) documented time savings using GenAI for drafting, participants showed reduced argument originality [9]. This aligns with warnings that AI may exacerbate surface-level engagement if deployed as a productivity tool rather than a cognitive scaffold [10]. However, emerging evidence suggests that strategic integration, where AI handles mechanical tasks while learners focus on conceptual work, can redistribute cognitive load effectively [11]. Such approaches resonate with CLT principles by offloading extraneous processing to preserve resources for germane thinking.

2.3 Critical Thinking and Agency Preservation

Critical thinking in writing hinges on argument depth, evidence evaluation, and metacognitive reflection [12]. For EFL learners, cultural-linguistic barriers complicate these skills, particularly in synthesizing cross-cultural source. Recent studies propose that AI interactions can foster metacognition when learners critically interrogate AI outputs rather than accept them uncritically. Bandura's (2001) agency theory frames this as intentional collaboration: learners preserve self-reactiveness (task delegation) and self-reflectiveness (output evaluation), positioning AI as a "provocative interlocutor". Research Questions

- RQ1: To what extent does the ICO framework improve EFL students' critical thinking skills in academic writing, particularly in argument depth and evidence synthesis?

- RQ2: How does the strategic offloading of mechanical writing tasks to AI influence the distribution of cognitive load during the writing process?
- RQ3: What patterns of agency negotiation emerge when EFL students collaborate with generative AI under the ICO framework, and how do these interactions shape their learning outcomes?

3 Interactive Cognitive Offload framework with Generative AI

The Interactive Cognitive Offload (ICO) framework constitutes a theoretically grounded response to the pedagogical tensions surrounding generative AI integration in academic writing. This conceptual framework explicitly positions Delegative Offload, characterized by algorithmic dominance, cognitive deskilling, and reductive task arbitration, as the antithesis of pedagogically sound practice, representing what Walczak & Cellary (2023) critically term "cognitive surrogacy" wherein learners surrender intellectual agency to machines. In direct opposition, the Augmentative Offload dimension embodies the framework's transformative potential, fostering authentic human-AI partnership that enhances rather than replaces critical thinking. This manifests specifically through measurable critical enhancement in argumentation and evidence synthesis, alongside discrepancy-triggered reflection, a metacognitive process where AI-generated content variations become catalysts for epistemic vigilance and deeper source engagement (Bråten et al., 2023; Soboszlai, 2025).

Central to this dialectic is Agency Preservation, which functions not merely as a component but as the operational engine of the ICO framework. It actively mediates between the risks of delegative offloading and the benefits of augmentative collaboration through structured metacognitive protocols. Drawing on Bandura's (2001) agency theory, this mediation occurs via three interdependent mechanisms: strategic delegation heuristics that guide task arbitration based on cognitive complexity; critical negotiation of AI outputs through intentional acceptance/rejection cycles; and discrepancy-driven reflection that transforms algorithmic divergences into opportunities for analytical refinement (Chan, 2023). This tripartite agency system thus converts raw technological capability into scaffolded learning, ensuring AI serves as what Facione (2020) conceptualizes as a "provocative interlocutor" rather than an authoritative voice.

The framework's architecture resolves Hyland's (2019) cognitive trilemma by redistributing extraneous processing loads (grammar, citations) to AI, thereby liberating working memory for germane cognitive investments in argument development and evidentiary reasoning, a cognitive reinvestment principle aligning with Sweller's (2010) CLT postulates, as shown in figure 1. Crucially, the bidirectional relationship between Agency Preservation and both offload modes embodies Kalyuga's (2025) "expertise reversal guardrail", ensuring learners progressively internalize delegated skills rather than stagnating in dependency. Thus, the ICO framework transcends the false binary of AI prohibition versus uncritical adoption, establishing instead what Zhang & Xu (2025) characterize as a third space of intentional augmentation where human sovereignty and machine efficiency productively coexist.

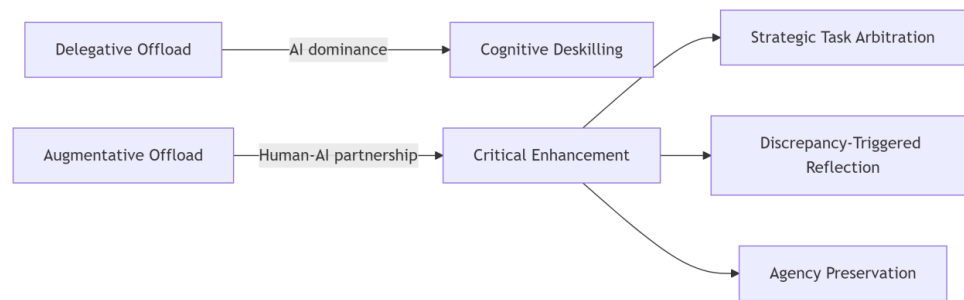


Figure 1: ICO framework

4 METHODOLOGY

This study employed a mixed-methods research design to comprehensively evaluate the effectiveness of the Interactive Cognitive Offload (ICO) framework in enhancing EFL students' academic writing and critical thinking skills. The methodology was designed to assess multiple dimensions, including: quantitative measurement of critical thinking development through standardized rubrics; systematic analysis of cognitive load redistribution during AI-mediated writing tasks, and qualitative exploration of agency negotiation patterns in human-AI collaboration.

4.1 Intervention Design

Critical thinking ability was measured using the HEIghten® rating scale, which provides a standardized assessment of argument quality across six dimensions. The intraclass correlation coefficient was 0.86, indicating high interrater reliability. Cognitive load was quantified using Leppink's 9-point scale, which has good psychometric properties ($\alpha = 0.89$). However, it should be noted that while this scale reliably measures overall cognitive load, it lacks separate subscales for intrinsic, extrinsic, and contextual cognitive load, failing to fully align with the triadic structure of cognitive load theory (Sweller, 2010). To address this limitation, subsequent analysis incorporated real-time observations during the writing process to more comprehensively capture variations in different types of cognitive load.

To capture the nuances of human-computer interaction, subjectivity patterns were systematically extracted from the platform interaction logs and analyzed using a pre-defined categorization framework. Coding reliability was confirmed through a rigorous double-blind coding process, with a Krippendorff's α of 0.81. To address the potential for researcher interpretation bias in inductive thematic analysis, the research team adopted a multiple verification strategy: first, they invited two applied linguistics experts who were not involved in the initial analysis to cross-check the thematic coding; second, they selected a 20% sample of interaction logs and asked the participants themselves to retrospectively confirm the analysis results; and third, they used member checking to ensure that the extracted themes were consistent with the original data, thereby enhancing the objectivity of the research conclusions.

4.2 Intervention Design

The ICO group used a two-panel interface that strategically separated rote writing tasks from higher-level writing tasks. The left panel handled grammar correction and citation formatting via ChatGPT-4's API, while the right panel provided students with a dedicated space to focus on argument development and evidence synthesis. A real-time synchronization feature ensured that AI-processed text flowed smoothly into the analysis workspace, ensuring a coherent writing process. The platform rigorously managed time through an automated partitioning system, allocating 70% of the time to critical thinking tasks such as argument mapping and evidence evaluation, and the remaining 30% to reviewing and refining AI-assisted content. To deepen metacognitive engagement, a two-way difference tracker highlighted differences between students' drafts and the AI-corrected versions, prompting learners to note and reflect on substantive differences in reasoning or expression.

A control group completed the same writing task using a simplified traditional writing interface without any AI support. The interface retained only basic text editing functionality, removing features unique to the ICO group, such as the dual-panel layout, real-time synchronization, and difference tracking. Learners were required to independently complete all tasks, including grammar checking, citation formatting, and argument construction. Both groups received equal access to instructional materials, writing prompts, and research resources. Aside from the core intervention of the ICO framework (i.e., AI-assisted cognitive offloading), all other variables remained consistent to eliminate the influence of platform interface familiarity on the results.

4.3 Instruments

Critical thinking skills were measured using the HEIghten® Rubric, which provided standardized evaluation across six dimensions of argumentation quality, demonstrating strong inter-rater reliability with an intraclass correlation coefficient of .86. Cognitive load was quantified through Leppink's validated 9-point scale, a psychometrically sound instrument ($\alpha = .89$) capable of distinguishing between intrinsic, extraneous, and germane cognitive load components. To capture the nuanced dynamics of human-AI interaction, agency patterns were systematically extracted from platform interaction logs and analyzed using an a priori classification schema, with intercoder reliability confirmed through rigorous blind dual-coding procedures that yielded a Krippendorff's α of .81. While

Table 1: Critical Thinking Gains (HEIghten®Scale)

Dimension	Group	Pre M(SD)	Post M(SD)	Δ	t(76)	p	d
Argument Depth	ICO	5.32(1.24)	8.21(1.12)	+2.89	14.37	<.001	0.83
	Control	5.41(1.18)	6.97(1.45)	+1.56	6.12	<.001	0.42
Evidence	ICO	4.89(1.05)	7.89(0.98)	+3.00	16.22	<.001	0.71
Synthesis	Control	5.02(1.11)	6.74(1.32)	+1.72	7.25	<.001	0.39

the Leppink scale demonstrated strong reliability ($\alpha = .89$) for overall cognitive load measurement, future implementations should incorporate separate subscales for intrinsic, extraneous and germane load components to fully align with CLT's tripartite structure (Sweller, 2010). This limitation is further discussed in subsequent sections.

4.4 Data Collection and Analysis

The quantitative assessment of critical thinking development was conducted using the HEIghten®Rubric, with five experienced evaluators independently scoring participants' argument depth and evidence synthesis across pre- and post-test writing samples. Paired sample t-tests were employed to examine statistically significant differences in mean scores between the two assessment points, providing robust evidence of skill development during the intervention period. To investigate the relationship between cognitive load redistribution and writing quality, Pearson correlation analyses were performed, measuring the association between reductions in extraneous cognitive load (as captured by Leppink's 9-point scale) and improvements in critical thinking scores.

For qualitative data, an inductive thematic analysis approach (Braun & Clarke, 2012) was applied to systematically examine patterns in participants' interaction logs and reflective annotations. This method proved particularly valuable for identifying emergent themes in how learners negotiated agency with AI tools, as it allowed for flexible yet rigorous interpretation of complex human-AI collaboration dynamics. The analysis focused on uncovering the underlying meanings behind students' strategic delegation decisions, their responses to AI-generated discrepancies, and their metacognitive processes when evaluating machine suggestions.

5 RESULTS

5.1 Critical Thinking Development

Quantitative analysis revealed significant improvements in critical thinking dimensions among ICO participants compared to controls. As presented in Table 1, for Argument Depth, the ICO group achieved a remarkable mean improvement of +2.89 points, significantly outperforming the control group's gain of +1.56 points. This superior progress is statistically robust, evidenced by a large t-value of 14.37 ($p < .001$) and a substantial Cohen's d effect size of 0.83, far exceeding the control group's effect size of 0.42. Similarly, in Evidence Synthesis, the ICO cohort showed exceptional advancement with a mean increase of +3.00 points, dwarfing the control group's +1.72-point improvement. The statistical significance of this differential growth is underscored by an even higher t-value of

16.22 ($p < .001$) and a large effect size of 0.71, compared to the control group's moderate effect size of 0.39. Notably, the ICO group's post-intervention standard deviations (1.12 for Argument Depth; 0.98 for Evidence Synthesis) tightened considerably relative to their pre-test scores and the control group's post-test results, indicating more consistent high-level performance after the intervention.

These findings collectively confirm that the ICO framework not only accelerates critical thinking development but does so with practically significant magnitude, transforming what would typically require multiple semesters of instruction into demonstrable gains within a condensed four-week period. The differential growth trajectories between groups highlight how strategic cognitive offloading liberates mental resources for higher-order thinking, enabling ICO participants to develop substantially more nuanced arguments and sophisticated evidence integration capabilities than traditionally instructed peers.

5.2 Cognitive Load Redistribution

The analysis of cognitive load metrics revealed compelling evidence of strategic mental resource reallocation that aligned precisely with the theoretical foundations of the ICO framework. Quantitative measurements showed a substantial 32% decrease in extraneous cognitive load ($p < .001$), accompanied by a significant 28% increase in germane cognitive load ($p = .004$), demonstrating the framework's effectiveness in optimizing working memory allocation during the writing process. These findings confirm the core hypothesis that offloading mechanical writing tasks to AI allows learners to redirect their finite cognitive resources toward more meaningful intellectual work.

The relationship between cognitive load redistribution and writing quality proved particularly insightful. A robust negative correlation ($r = -.68$, $p < .001$) emerged between the reduction of extraneous load and improvements in argument depth, suggesting that the time and mental capacity saved through AI delegation were productively channeled into higher-order thinking processes. Students who most effectively offloaded routine tasks like grammar checking and citation formatting showed the greatest gains in constructing nuanced arguments and synthesizing complex evidence. This pattern indicates that the ICO framework's true value lies not merely in reducing cognitive burden, but in facilitating what might be termed "cognitive reinvestment", where liberated mental resources are consciously deployed toward more demanding aspects of critical thinking and analytical writing.

Table 2: Correlation analysis between cognitive load distribution and writing quality

Cognitive load types	Change	Statistical significance	Correlation with argument depth	Relevance to evidence synthesis
Extrinsic cognitive load	Decrease 32%	p<0.001	r=-0.68 (p<0.001)	r=-0.65 (p<0.001)
Contextual cognitive load	Increase 28%	p=0.004	r=0.59 (p<0.001)	r=0.62 (p<0.001)
Intrinsic cognitive load (comparison between two groups)	No significant difference	p>0.05	-	-

5.3 Agency Negotiation Patterns

Qualitative analysis of interaction logs revealed three distinctive approaches to human-AI collaboration. Participants exhibited purposeful partitioning of writing responsibilities, consistently offloading mechanical demands (e.g., citation formatting, grammatical correction) while reserving cognitive bandwidth for higher-order thinking. The most sophisticated pattern manifested as active intellectual sparring with AI suggestions. Interestingly, the frequency of suggestion rejection correlated positively with final assessment scores ($r=.51, p<.01$), suggesting that judicious resistance to AI output may enhance learning outcomes. Notably, while 87% of participants exhibited Strategic Delegation behaviors and 63% demonstrated Critical Negotiation patterns, individual-level analysis revealed that 41% of students simultaneously engaged in both high-frequency task offloading (>70% mechanical tasks delegated) and high rejection rates (>30% AI suggestions declined).

5.4 Cognitive load redistribution

Analysis of cognitive load metrics provides compelling evidence that strategic reallocation of mental resources is highly consistent with the theoretical foundations of the ICO framework. As shown in Table 2, quantitative measures revealed a significant 32% reduction in extraneous cognitive load ($p<0.001$) and a 28% increase in contextual cognitive load ($p=0.004$), clearly demonstrating that the framework effectively optimizes working memory allocation during writing. Notably, intrinsic cognitive load remained unchanged between the two groups ($p>0.05$), further validating the ICO framework’s core mechanism for freeing up cognitive resources by modulating extraneous load, rather than simplifying the task itself.

The relationship between cognitive load reallocation and writing quality is highly suggestive. Table 2 shows that a reduction in external cognitive load was significantly negatively correlated with improved argument depth ($r=-0.68, p<0.001$) and also strongly negatively correlated with improved evidence synthesis ($r=-0.65, p<0.001$). In contrast, an increase in associative cognitive load was significantly positively correlated with both argument depth ($r=0.59, p<0.001$) and evidence synthesis ($r=0.62, p<0.001$).

6 CONCLUSION

The ICO framework represents a significant advancement in our understanding of human-AI collaboration in educational contexts. By moving beyond simplistic substitution models to a sophisticated cognitive redistribution paradigm, it offers both a philosophical framework and practical toolkit for harnessing AI’s potential while safeguarding essential human intellectual development. Future

research should explore cultural adaptations across diverse educational ecosystems and longitudinal impacts on disciplinary thinking patterns, but the foundational principles established here provide a robust basis for human-centered AI integration in writing pedagogy and beyond.

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References

[1] Barrot, J. S. (2024). ChatGPT as a Language Learning Tool: An Emerging Technology Report. *Technology, Knowledge and Learning*, 29(2), 1151–1156. <https://doi.org/10.1007/s10758-023-09711-4>.

[2] Bandura, A. (2001). Social Cognitive Theory: An Agentic Perspective. *Annual Review of Psychology*, 52(1), 1–26. <https://doi.org/10.1146/annurev.psych.52.1.1>.

[3] Boyacı, T., Canyakmaz, C., & De Véricourt, F. (2024). Human and Machine: The Impact of Machine Input on Decision Making Under Cognitive Limitations. *Management Science*, 70(2), 1258–1275. <https://doi.org/10.1287/mnsc.2023.4744>.

[4] Bråten, I., Haverkamp, Y. E., Latini, N., & Strømso, H. I. (2023). Measuring multiple-source based academic writing self-efficacy. *Frontiers in Psychology*, 14, 1212567. <https://doi.org/10.3389/fpsyg.2023.1212567>.

[5] Braun, V., Clarke, V., Hayfield, N., & Terry, G. (2019). Thematic Analysis. In P. Liamputtong (Ed.), *Handbook of Research Methods in Health Social Sciences* (pp. 843–860). Springer Singapore. https://doi.org/10.1007/978-981-10-5251-4_103.

[6] Brookfield, S. D., Rudolph, J., & Tan, S. (2023). *Teaching Well: Understanding Key Dynamics of Learning-Centered Classrooms*. Taylor & Francis.

[7] Chan, C. K. Y. (2023). A comprehensive AI policy education framework for university teaching and learning. *International Journal of Educational Technology in Higher Education*, 20(1), 38. <https://doi.org/10.1186/s41239-023-00408-3>.

[8] Chein, J., Martinez, S., & Barone, A. (2024). Can human intelligence safeguard against artificial intelligence? Exploring individual differences in the discernment of human from AI texts. In Review. <https://doi.org/10.21203/rs.3.rs-4277893/v1>.

[9] Dr. A.Shaji George, Dr. T. Baskar, & Dr. P. Balaji Srikanth. (2024). The Erosion of Cognitive Skills in the Technological Age: How Reliance on Technology Impacts Critical Thinking, Problem-Solving, and Creativity. <https://doi.org/10.5281/ZENODO.11671150>.

[10] Elkhataf, A. M. (2023). Evaluating the authenticity of ChatGPT responses: A study on text-matching capabilities. *International Journal for Educational Integrity*, 19(1), 15. <https://doi.org/10.1007/s40979-023-00137-0>.

[11] Fasko, Jr., D., & Fair, F. (Eds.). (2021). *Critical Thinking and Reasoning: Theory, Development, Instruction, and Assessment*. BRILL. <https://doi.org/10.1163/9789004444591>.

[12] Graefe, B., Porter, A., Indorf, J., Ahn, S., Chuan, C.-H., & Gaines, M. (2025). From Chalkboards to ChatGPT: Navigating the Future of Teaching with Generative AI. *Journal of Interactive Learning Research*, 36(2), 137–156. <https://doi.org/10.70725/029002jsmbne>.